08 - Model Evaluation

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Confusion matrix

For this project, churn is the positive class. TP is a churner correctly predicted as churn; FN is a churner predicted as non-churn; FP is a non-churner predicted as churn; TN is a non-churner correctly predicted as non-churn.

	Predicted churn	Predicted non-churn
Actual churn	TP	FN
Actual non-churn	FP	TN

2. Evaluation metrics

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Accuracy is the fraction of all predictions that are correct. Precision answers: among customers flagged as churners, what fraction actually churned? Recall answers: among all actual churners, what fraction did we identify? F1 is the harmonic mean of precision and recall.

3. Why recall and precision matter

A false negative is a churner missed by the model. If the business does not contact that customer, it loses a potential retention opportunity. A false positive is a customer flagged as at risk who would not have churned; outreach may waste campaign budget or annoy the customer. A sensible decision threshold balances these costs.

4. Reported baseline results

Model (original class distribution)	Churn precision	Churn recall	Churn F1	Accuracy
Decision Tree	0.63	0.49	0.55	about 0.78
Random Forest	0.69	0.45	0.55	about 0.80

The model notebook reports 1,407 test observations: 1,023 non-churn and 384 churn. Random Forest has higher churn precision but lower churn recall than Decision Tree in this baseline comparison. This illustrates why accuracy alone is inadequate.

5. Reported results after SMOTEENN

Configuration	Churn precision	Churn recall	Churn F1	Reported accuracy
Decision Tree + SMOTEENN	0.91	0.98	0.94	0.9344
Random Forest +	0.94	0.96	0.95	0.9427

SMOTEENN				
Random Forest + SMOTEENN + PCA	0.72	0.81	0.77	0.7239

Important: The post-SMOTEENN test results are not valid estimates of generalization because the repository resamples the full dataset before splitting it. Present them only as reported notebook outputs and explicitly qualify them.

6. Confusion matrices reported

Decision Tree + SMOTEENN: [[477, 63], [14, 620]]. Random Forest + SMOTEENN: [[478, 40], [27, 625]]. In scikit-learn ordering, rows are actual classes and columns are predicted classes.

7. PCA experiment

The repository applies PCA(n_components=0.9) to the resampled training data, fits PCA on training data and transforms the corresponding test data. The PCA Random Forest result is reported as approximately 72.39% accuracy, with churn precision 0.72, recall 0.81 and F1 0.77. The author therefore retains the Random Forest without PCA.

PCA projects features into orthogonal linear combinations called principal components, ordered by explained variance. Retaining 90% variance does not guarantee retaining the information most useful for classification. PCA can also make tree-based splits less directly useful because the components mix original variables.

8. Additional metrics worth knowing

- ROC-AUC evaluates ranking across true-positive and false-positive rates.
- PR-AUC / average precision summarizes precision-recall behaviour and is often informative when the positive class is uncommon.
- Calibration checks whether predicted probabilities correspond to observed event frequencies.
- A confusion matrix should be evaluated at the chosen operating threshold.

9. Interview-ready explanation

Initially, the Decision Tree and Random Forest showed weak recall for churners, around 49% and 45% respectively. The notebook then explored SMOTEENN and reported much higher scores, with Random Forest reaching approximately 94% accuracy and 96% churn recall. However, because SMOTEENN was applied before the train-test split, those resampled test figures are subject to leakage and should not be treated as reliable unseen-data performance. I would rerun evaluation with stratified splitting and resampling confined to training folds.

10. Questions to prepare

Q. Why can accuracy be misleading?

A majority-class model can score around the majority prevalence while missing the minority class entirely.

Q. What is the precision-recall trade-off?

Changing the threshold typically changes both: flagging more observations tends to increase recall but can reduce precision.

Q. Why PR-AUC for an imbalanced task?

It focuses on precision and recall of the positive class and is often more informative than ROC-AUC when the positive class is rare.

Q. What does PCA(n_components=0.9) mean?

Retain enough components to explain at least 90% of the variance in the data used to fit PCA.

Q. Why can PCA reduce tree performance?

Linear combinations can obscure simple axis-aligned threshold patterns, and variance preservation does not equal predictive-information preservation.